

$$1.a) I(p^\dagger)^{-1} = (p^{-1})^\dagger$$

$$(p^\dagger)^{-1} p^\dagger = (p^{-1})^\dagger p^\dagger = I$$

$$I = (p p^{-1})^\dagger = I^\dagger = I$$

$$II. (p q)^{-1} = q^{-1} p^{-1}$$

$$I = (p q)^{-1} p q = q^{-1} (p^{-1} p) q = q^{-1} I q = I$$

$$III. Si [p, q] = 0, entonces p q^{-1} = q^{-1} p$$

$$p q = q p$$

$$q^{-1} p q = q^{-1} q p = p$$

$$q^{-1} p q q^{-1} = p q^{-1}$$

$$q^{-1} p = p q^{-1}$$

$$IV. (e^p)^\dagger = e^{p^\dagger}$$

$$(e^p)^\dagger = \sum_{n=0}^{\infty} \frac{(p^n)^\dagger}{n!} = \sum_{n=0}^{\infty} \frac{(p^\dagger)^n}{n!} = e^{p^\dagger}$$

$$V. p e^q p^{-1} = e^{p q p^{-1}}$$

$$p e^q p^{-1} = p \left(\sum_{n=0}^{\infty} \frac{q^n}{n!} \right) p^{-1} = \sum_{n=0}^{\infty} \frac{p q^n p^{-1}}{n!} = \sum_{n=0}^{\infty} \frac{(p q p^{-1})^n}{n!} = e^{p q p^{-1}}$$

$$b) \tilde{A}^\dagger = (U^{-1} A U)^\dagger = U^\dagger A^\dagger (U^{-1})^\dagger$$

$$\tilde{A}^\dagger = U^{-1} A U = \tilde{A}$$

$$c) V = e^{iA}$$

$$V^\dagger = (e^{iA})^\dagger = e^{(iA)^\dagger} = e^{-iA} = V^{-1}, V V^{-1} = I$$

$$d) \tilde{K}^\dagger = (u^{-1} K u)^\dagger = u^\dagger K^\dagger (u^{-1})^\dagger = u^\dagger (-K) u = -u^\dagger K u = -\tilde{K}$$

$$\text{Para } K = iA, (iA)^\dagger = -iA$$

$$e) (AB)^\dagger = B^\dagger A^\dagger = BA = AB \text{ si y solo si } BA = AB$$

$$f) I. (I-S)(I+S) = I-S^2 = (I+S)(I-S)$$

$$II. [(I-S)(I+S)]^T = (I+S)^T (I-S)^T = (I-S)(I+S) \xrightarrow{\text{Simétrica}} (I-S)(I+S)$$

$$\text{Para la ortogonalidad, } M^T M = I:$$

$$M^T = ((I+S)^{-1})^T (I-S)^T = ((I+S)^T)^{-1} (I-S)^T = (I-S)^{-1} (I+S)$$

$$M^T M = (I-S)^{-1}(I+S)(I-S)(I+S)^{-1} = (I-S)^{-1}(I-S)(I+S)(I+S)^{-1} = I$$

$$g) R = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}, R = (I-S)(I+S)^{-1}$$

$$R(I+S) = (I-S) \Rightarrow R + RS = I - S \Rightarrow RS + S = I - R$$

$$(R+I)S = I - R \Rightarrow S = (R+I)^{-1}(I-R)$$

$$S = \begin{pmatrix} 0 & -\frac{\sin \theta}{1 + \cos \theta} \\ \frac{\sin \theta}{1 + \cos \theta} & 0 \end{pmatrix} = \tan \frac{\theta}{2} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

2. a) En la base monomial:

$$[f]_{B_1} = \begin{bmatrix} 0 \\ 5 \\ 3 \\ 4 \\ 0 \end{bmatrix}$$

En base Legendre:

$$f(t) = a_0(1) + a_1(t) + a_2\left(\frac{1}{2}(3t^2-1)\right) + a_3\left(\frac{1}{2}(5t^3-3t)\right) + a_4\left(\frac{1}{8}(35t^4-30t^2+3)\right) \\ = \underbrace{(a_0 - \frac{1}{2}a_2 + \frac{3}{8}a_4)}_0 \cdot 1 + \underbrace{(a_1 - \frac{3}{2}a_3)}_5 \cdot t + \underbrace{(\frac{3}{2}a_2 - \frac{15}{4}a_4)}_3 \cdot t^2 + \underbrace{(\frac{5}{2}a_3)}_4 \cdot t^3 + \underbrace{\frac{35}{8}a_4}_0 \cdot t^4$$

$$a_4 = 0, a_3 = \frac{8}{5}, a_2 = 2, a_1 = \frac{37}{5}, a_0 = 1$$

$$[f]_{B_2} = \begin{pmatrix} 1 \\ 37/5 \\ 2 \\ 8/5 \\ 0 \end{pmatrix}$$

La matriz P transforma de Legendre a monomial, P^{-1} al revés:

$$P = \begin{pmatrix} 1 & 0 & -\frac{1}{2} & 0 & \frac{3}{8} \\ 0 & 1 & 0 & -\frac{3}{2} & 0 \\ 0 & 0 & \frac{3}{2} & 0 & -\frac{15}{4} \\ 0 & 0 & 0 & \frac{5}{2} & 0 \\ 0 & 0 & 0 & 0 & \frac{35}{8} \end{pmatrix}, P^{-1} = \begin{pmatrix} 1 & 0 & \frac{1}{3} & 0 & \frac{1}{6} \\ 0 & 1 & 0 & \frac{2}{5} & 0 \\ 0 & 0 & \frac{2}{3} & 0 & \frac{4}{15} \\ 0 & 0 & 0 & \frac{2}{5} & 0 \\ 0 & 0 & 0 & 0 & \frac{8}{35} \end{pmatrix}$$

b) Construyo el proyector usando Legendre por ser ortogonales:

$$g_2(f) = \sum_{n=0}^2 \frac{\langle f, P_n \rangle}{\langle P_n, P_n \rangle} P_n \Rightarrow \langle P_n, P_n \rangle = \frac{2}{2n+1}$$

$$\langle f, P_0 \rangle = \int_{-1}^1 f(t) \cdot 1 dt = 2, a_0 = \frac{\langle f, P_0 \rangle}{\langle P_0, P_0 \rangle} = \frac{2}{2/1} = 1$$

$$\langle f, p_1 \rangle = \int_{-1}^1 f(t) t dt = \frac{74}{15}, a_1 = \frac{\langle f, p_1 \rangle}{\langle p_1, p_1 \rangle} = \frac{74/15}{2/3} = \frac{37}{5}$$

$$\langle f, p_2 \rangle = \int_{-1}^1 f(t) \frac{1}{2}(3t^2 - 1) dt = \frac{4}{5}, a_2 = \frac{\langle f, p_2 \rangle}{\langle p_2, p_2 \rangle} = \frac{4/5}{2/5} = 2$$

La proyección es $g_2(f) = 1 \cdot p_0 + \frac{37}{5} p_1 + 2 p_2$
y en base monomial:

$$g_2(f) = 1 + \frac{37}{5} t + 2 \left(\frac{1}{2}(3t^2 - 1) \right) = \frac{37}{5} t + 3t^2$$

La proyección ortogonal es única como polinomio, aunque su representación numérica cambia al usar diferentes bases

$$c) (e^0 f)(t) = \sum_{k=0}^{\infty} \frac{1}{k!} D^k f(t) = \sum_{k=0}^{\infty} \frac{1}{k!} f^{(k)}(t) = f(t+1)$$

El inverso de T es la traslación opuesta:

$$T^{-1} f(t) = f(t-1) = e^{-D} f(t)$$

En la base monomial la matriz T^{-1} se obtiene expandiendo $(t-1)^n$:

$$\left. \begin{aligned} (t-1)^0 &= 1 \\ (t-1)^1 &= t-1 \\ (t-1)^2 &= t^2-2t+1 \\ (t-1)^3 &= t^3-3t^2+3t-1 \\ (t-1)^4 &= t^4-4t^3+6t^2-4t+1 \end{aligned} \right\} T^{-1} = \begin{pmatrix} 1 & -1 & 1 & -1 & 1 \\ 0 & 1 & -2 & 3 & -4 \\ 0 & 0 & 1 & -3 & 6 \\ 0 & 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

y hallando $(T^{-1})^{-1} = T$:

$$T = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 2 & 4 \\ 0 & 0 & 1 & 3 & 6 \\ 0 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \Rightarrow T^{\dagger} = T^T = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 \\ 1 & 3 & 3 & 1 & 0 \\ 1 & 4 & 6 & 4 & 1 \end{pmatrix}$$

T no es hermítico porque $T^{\dagger} \neq T$ y no es unitario porque $T^{\dagger} \neq T^{-1}$.

d)

$$M_{\text{mono}}(T) = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 2 & 4 \\ 0 & 0 & 1 & 3 & 6 \\ 0 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \det(M_{\text{mono}}) = 1, \text{Tr}(M_{\text{mono}}) = 5$$

En base de Legendre:

$$M_L(T) = P^{-1} M_{\text{mono}} P = \begin{pmatrix} 1 & 1 & \frac{3}{2} & \frac{7}{2} & \frac{75}{8} \\ 0 & 1 & 3 & \frac{15}{2} & \frac{41}{2} \\ 0 & 0 & 1 & 5 & \frac{35}{2} \\ 0 & 0 & 0 & 1 & \frac{7}{2} \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \det(M_L) = 1, \text{tr}(M_L) = 5$$

e) Ya hallé para la base monomial, para Legendre uso $M_L(T)$:

$$(M_L(T))^{-1} = \begin{pmatrix} 1 & -1 & \frac{3}{2} & -\frac{7}{2} & \frac{75}{8} \\ 0 & 1 & -3 & \frac{15}{2} & -\frac{41}{2} \\ 0 & 0 & 1 & -5 & \frac{35}{2} \\ 0 & 0 & 0 & 1 & -\frac{7}{2} \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}, M_L(T^+) = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ \frac{3}{2} & 3 & 1 & 0 & 0 \\ \frac{7}{2} & \frac{15}{2} & 5 & 1 & 0 \\ \frac{75}{8} & \frac{41}{2} & \frac{35}{2} & 7 & 1 \end{pmatrix}$$

4. a) $(A+B)^2 = (A+B)(A+B) = A^2 + AB + BA + B^2 = A^2 + 2AB + B^2$

Si: $AB \neq BA$

$$(A+B)^2 = A^2 + AB + BA + B^2$$

b) $(A+B)^3 = (A+B)(A+B)^2 = A^3 + A^2B + ABA + AB^2 + BA^2 + BAB + B^2A + B^3$
 $= A^3 + 3A^2B + 3B^2A + B^3$

Si: $AB \neq BA$

$$(A+B)^3 = A^3 + A^2B + ABA + AB^2 + BA^2 + BAB + B^2A + B^3$$

5. Primer caso:

$$L|y\rangle = L(L_+|x\rangle) = (L - L_+)(L_+|x\rangle)$$

$$[L_-, L_+] = L_-L_+ - L_+L_- = I \Rightarrow L_-L_+ = L_+L_- + I$$

$$L|y\rangle = (L_+L_- + I)(L_+|x\rangle) = L_+(L_-L_+|x\rangle) + L_+|x\rangle$$

$$= L_+\lambda|x\rangle + L_+|x\rangle = (\lambda+1)L_+|x\rangle = (\lambda+1)|y\rangle$$

Segundo caso:

$$L|z\rangle = L(L_-|x\rangle) = (L - L_+)(L_-|x\rangle)$$

$$[L_-, L_+] = L_-L_+ - L_+L_- = I \Rightarrow L_+L_- = L_-L_+ - I = L_- - I$$

$$L|z\rangle = L_-(L_-I)|x\rangle = L_-(L_-|x\rangle - |x\rangle) = L_-(\lambda|x\rangle - |x\rangle)$$

$$= (\lambda-1)L_-|x\rangle = (\lambda-1)|z\rangle$$